The complex Grothendieck constant

New bounds, machine-checked in Lean

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https://github.com/agalatan/grothendieck-lean

October 2026

$$1.401691989174 \;<\; K_G^{\mathbb{C}} \;<\; 1.404599424$$

Both bounds are theorems of a Lean 4/Mathlib development, each resting on one explicitly stated family of externally certified numerical inputs (Section 3).

Abstract

Grothendieck's inequality bounds how much a semidefinite (vector) relaxation of a bilinear optimisation problem can exceed the true optimum. The best constant in the complex case, $K_G^{\mathbb{C}}$, has been known since 1987 to lie roughly in $[1.338, 1.405]$; its first two decimals were settled only in September 2026. We improve both bounds, to an interval of width 0.0029, and verify the arguments in Lean. This note explains the problem, the two constructions, exactly what the computer checks, and why the remaining gap is hard: the two sides are dual optimisation problems, and the obstruction on both is the nonlinear (cubic and higher) structure of bounded functions of Gaussian vectors.

1 The question

Let $A = (a_{ij})$ be an $m \times n$ complex matrix. Compare

$$C(A) = \sup_{|\varepsilon_i|=|\delta_j|=1} \Big|\sum_{i,j} a_{ij}\,\overline{\varepsilon_i}\,\delta_j\Big|, \qquad V(A) = \sup_{\|x_i\|=\|y_j\|=1} \Big|\sum_{i,j} a_{ij}\,\langle x_i, y_j\rangle\Big|,$$

the first over unimodular scalars, the second over unit vectors in any complex Hilbert space. Clearly $C(A) \le V(A)$. Grothendieck proved in 1953 that $V(A) \le K\,C(A)$ for a constant K independent of m, n and the dimension. The smallest such K is the *complex Grothendieck constant* $K_G^{\mathbb{C}}$.

Why it matters.

- *Optimisation.* $V(A)$ is a semidefinite program; $C(A)$ is a hard combinatorial problem. $K_G^{\mathbb{C}}$ is the best approximation ratio of the semidefinite relaxation, and rounding algorithms for it are exactly upper-bound proofs.
- *Functional analysis.* K_G is the norm of the identity between the injective and projective tensor norms on $\ell_\infty \otimes \ell_\infty$, and controls absolutely summing operators (Grothendieck's original setting).
- *Quantum information.* The real constant $K_G^{\mathbb{R}}$ is the largest ratio between quantum and classical values of two-player XOR games (Tsirelson); the finite-order constants $K_G(d)$ govern the nonlocality of qubit states.

History.

year	who	lower bound	upper bound
1953	Grothendieck		$K < \infty$
1984	Davie	1.33807	
1987	Haagerup		1.40490913… (Gaussian phase rounding)
Sep 2026	Guo, Fang, Lu	1.35584631827	
Sep 2026	Heilman, Jones, Malavolta	1.4	1.404898554746
Oct 2026	**this work**	**1.401691989174**	**1.404599424**

Haagerup's integral $1/\gamma^\dagger = 1.404575934\ldots$, $\gamma^\dagger = \int_0^{\pi/2} \cos^2 t\,(1+\sin^2 t)^{-1/2} dt$, has been proposed as the exact value. It lies inside our interval, $2.35 \cdot 10^{-5}$ below the upper bound. For the real constant the corresponding recent results are $1.773 \le K_G^{\mathbb{R}} \le 1.7799$ (Heilman–Jones–Malavolta, October 2026).

2 Two dual constructions

Both bounds use the same Gaussian picture. Let $Z = (z_1, \ldots, z_n)$ be a standard complex Gaussian vector and decompose $L^2(\mathbb{C}^n, \gamma)$ into the *Wick chaoses* $H_{p,q}$ (polynomials of bidegree (p, q) in $z, \bar{z}$, orthogonalised). Under the rotation $z \mapsto e^{i\theta} z$ every relevant object has *charge one*, so only the bidegrees $(1,0), (2,1), (3,2), \ldots$ (degrees $1, 3, 5, \ldots$) appear.

2.1 Upper bounds: rounding schemes

Given unit vectors, project a Gaussian onto them and round: $\varepsilon_i = f(\langle G, x_i\rangle)$ with $|f| \le 1$. The correlation of two rounded values depends only on $t = \langle x_i, y_j\rangle$,

$$\mathbb{E}\big[\overline{f(\cdot)}\, g(\cdot)\big] = H(t) = \sum_{k \ge 0} c_k\, t|t|^{2k}.$$

If $H(t) = c\,t + r(t)$ with $\|r\|_A = \sum_k |c_k - c\,\delta_{k0}| < c$, a tensor-power argument (Krivine's trick) gives $K_G^{\mathbb{C}} \le 1/(c - \|r\|_A)$. Haagerup's choice $f = \text{phase}$ gives 1.40491. Our rounding is a smoothed two-coordinate phase rounding, $\text{phase}\big(z + \sqrt{1+a^2}\, v(|z|^2)\, w\big)$ with a twelve-term Laguerre profile v, multiplied by a damped polynomial phase $e^{i\varphi}$; its kernel is $H(t) = \sum E[n,m]\, p(t)^n (-\tau(t))^m$ with an odd polynomial p of degree 101 and $\tau(t) = (t^3 + a^2 t^7)/(1+a^2)$. Compared with $c\,t$, $c = 625000/877873$, the residual is at most $133 \cdot 10^{-8}$:

$$K_G^{\mathbb{C}} \le \frac{1}{c - 133 \cdot 10^{-8}} = \frac{87787300000000}{62499883242891} < 1.404599424.$$

The whole difficulty is to keep *every* coefficient c_k, $k \ge 1$, small at once: improving the cubic coefficient c_1 tends to spoil $c_2, c_3, \ldots$, and the infinite tail has to be bounded too.

2.2 Lower bounds: Hermite projection games

A lower bound needs an operator T whose *vector* value is close to 1 while its *scalar* value is small. We use

$$T = P_{1,0} - \epsilon\, P_{2,1}, \qquad \epsilon = \tfrac{149}{200},$$

the projection onto linear functions minus a multiple of the projection onto the cubic chaos.

- *Vector value.* With $u_\ell = z_\ell/\|z\|$, $\sum_\ell \langle u_\ell, T u_\ell\rangle \ge 1 - (1+\epsilon)/(n+1) \to 1$.
- *Scalar value.* For all $|f|, |g| \le 1$, $|\langle f, Tg\rangle| \le B = 0.713423496548$.

A finite-partition transfer turns this into matrices, so $K_G^{\mathbb{C}} \geq 1/B > 1.401691989174$.

The scalar cap is the hard part. Writing $h = \frac{1}{2}(f+g)$, $k = \frac{1}{2}(f-g)$,

$$\Re\langle f, Tg\rangle = t^2 - \epsilon\|P_{2,1}h\|^2 - \|P_{1,0}k\|^2 + \epsilon\|P_{2,1}k\|^2, \qquad t = \|P_{1,0}h\|.$$

Only the last term is positive: we must show that a *bounded* function cannot have too large a cubic part without paying elsewhere. After a rotation $P_{1,0}h = t\,z_1$, and the cubic part of k splits into pieces that involve z_1 (*visible* sectors) and a *remote* residual $q \in H_{2,1}$ in the other coordinates. The proof then has three layers.

1. **Row reduction.** For each of 19 profiles r, the contribution of the visible sectors is the supremum of an explicit finite-dimensional Gaussian integral, a convex one-variable *potential* Ψ_r, evaluated at the scale of the remote residual. One gate variable is shared across rows.
2. **The law of the remote cubic.** What remains is a statement about the distribution of $U = |q|^2/\mathbb{E}|q|^2$ for an arbitrary cubic Wick polynomial in arbitrarily many variables. We use exact identities and inequalities valid for *every* such cubic: the fourth-moment cap $\mathbb{E}U^2 \leq 62$; the sixth-moment bound $\mathbb{E}U^3 \leq 9m - 12 + 40.45313\,(m-2)^{3/2}$, $m = \mathbb{E}U^2$, which rests on thirteen tensor-contraction facts including the Wick expansion of the sixth cumulant; a first-moment bound; and families of *Stein cuts* $\mathbb{E}\,G(U) \geq 0$, obtained by Gaussian integration by parts and completing squares against spectral budgets, a covariance order and *contraction-Gram budgets* (positive semidefinite Gram matrices of the coefficient tensor).
3. **A finite certificate.** The parameters are covered by 7,875 boxes in (t, τ, x); on each box a positive combination of rows and node certificates (336 joint nodes, 9,237 row nodes, each a polynomial majorant checked in exact rational arithmetic) bounds the scalar value by B.

2.3 The two problems are one

For the real constant, Heilman, Jones and Malavolta proved an exact minimax: over rounding schemes μ with kernel coefficients $c_m(\mu)$ and Hermite projection games $Q = \sum_m q_m \Pi_m$,

$$\frac{1}{K_G} = \sup_{\mu}\Big[c_1(\mu) - \sum_{m\geq 3}|c_m(\mu)|\Big] = \inf_{Q}\ \mathrm{OPT}(Q),$$

and the same duality holds in the complex case on charge-one bidegrees. An upper bound is a good point of the convex set of achievable kernels; a lower bound is a supporting hyperplane of that set. There is *no intrinsic gap*: the gap between our bounds is entirely the gap between restricted constructions and the cost of certifying them.

3 What exactly the computer checks

Proved in Lean	**Assumed (certified outside Lean)**
Lower bound. `lower_bound_of_nodeCaps : NodeCaps → IsGrothendieckBound K → 1.401691989174 < K` the witness T on L^2 of the Gaussian measure and its vector value; the transfer to matrices; the polarisation identity; the per-row Gaussian reduction of $\langle f, Tg\rangle$ for every unimodular pair (frames, fibre decomposition, Parseval, gates); all moment, Stein, spectral and contraction-Gram inequalities for every circular cubic, including the sixth-cumulant Wick identity; the finite certificate (one `native_decide`)	`NodeCaps`: the 19 potentials Ψ_r (defined in Lean) lie below their stored rational caps at the tabulated nodes. These caps come from exact-arithmetic certificates (interval integration and anchored quadratic majorants), replayed independently but not yet formalised.
Upper bound. `complex_grothendieck_upper : KernelHyps E EX E0 → IsGrothendieckBound (87787300000000/62499883242891)` the transfer inequality (tensor powers); kernel composition and locality; the head of the kernel through degree 501 from 21,168 stored energies by a verified interval evaluator (one `native_decide`); the tail bound; the endpoint	`KernelHyps`: the stored tables are the energies of the rounding (validity, summability, parity); the comparison between the rounding and its truncation (93.9% of the residual budget); a weighted square-sum bound for the tail (5.4%); two table budgets. The part computed in Lean is 0.54%.

Axioms: `propext`, `Classical.choice`, `Quot.sound`, and exactly two pinned `native_decide` facts. No `sorry`; CI builds everything and checks the axiom report (140 theorems). The development has about 28,000 lines of hand-written Lean and 11,000 lines of generated certificate data. The lower bound is close to fully formal: only one-variable numerical caps remain. The upper bound still rests mainly on an external analytic comparison lemma.

4 Why the gap is hard

The witness class caps the lower bound. A game supported on degrees $\{1, 3\}$ sees only the cubic constraint on roundings. Even a perfect evaluation of the norm of $P_{1,0} - \epsilon P_{2,1}$ cannot certify more than about 1.40226 (numerical, best $\epsilon \approx 0.756$). Proved obstructions are known for higher degrees: every game of degree ≤ 5 is capped at 1.4036705, and every game of degree ≤ 7 at 1.4044895, both below Haagerup's value. Reaching the upper end of the interval therefore needs games of degree ≥ 9; in the real case the certified game uses all odd degrees up to 21.

Certifying higher degree is the real bottleneck. Two different certificate architectures, gated rows and support functions, both lose about $2 \cdot 10^{-2}$ on the best degree-5 game. The reason is structural: a bounded function with a large cubic part is forced to have a degree-five part too. Through the visible coordinate, the cubic part sees the average of the remote coefficient profile and the quintic part its variation, and boundedness makes the profile vary. A certificate that treats the chaos degrees as independent cannot see this coupling.

Why the complex case is harder than the real one. For real ± 1 responses, $h = \frac{1}{2}(f + g)$ and $k = \frac{1}{2}(f - g)$ have disjoint supports, so the two sides of a game decouple at every degree. For complex unimodular responses $|h|^2 + |k|^2 = 1$ and $\Re(h\bar{k}) = 0$ force $f = e^{i\alpha}e^{i\theta}$, $g = e^{i\alpha}e^{-i\theta}$: one shared phase field α drives both sides, and they never separate. The same shared phase is why

marginal laws do not determine the relevant nonlinear correlations: $(H(X), X)$ and $(H(Y), X)$, with $H(X) = X(|X|^2 - 2)/\sqrt{2}$ and X, Y independent, have identical marginals and covariances, but $\mathbb{E}[\bar{r}\, q|q|^2]$ equals $8\sqrt{2}$ in the first case and 0 in the second.

5 Open problems

1. Formalise the node caps (`NodeCaps`), which would make the lower bound fully machine-checked.
2. Formalise the rounding behind the upper bound: its Hermite expansion, the comparison lemma and the contour tail bound.
3. Certify a game of degree ≥ 5 with a *multi-grade fibre*: one bounded function of the visible coordinate and one remote direction per degree, so that the cross-degree coupling is kept. This is the route to a lower bound above 1.403.
4. Run a complex primal/dual search, alternating linear programs over games of degree $\leq D$ and pools of roundings, to locate $K_G^{\mathbb{C}}$ to about 10^{-4} and to test whether Haagerup's value $1/\gamma^{\dagger}$ is the answer.
5. Prove the sharp first-moment bound $\mathbb{E}|q| \leq \frac{\sqrt{\pi}}{2}\|q\|_2$ for cubic Wick polynomials. It is natural, it appears to be open, and it is the largest single lever on the law side.

How this was made

The constructions and certificates were found by several AI research systems working on a shared manuscript, cross-checking each other's packages by independent replay. This Lean development turns the certificates into theorems: every claim above marked as proved is a Lean theorem whose axioms are checked in CI. The research manuscript and the certificate packages are kept outside the repository.

References

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